BT3041 Analysis and Interpretation of Biological Project Data: Report

Cold vs Hot Tumour Classification from Bulk RNA-seq Using a Multi-Statistical Pipeline

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Abstract

This term-paper project develops and validates a complete end-to-end framework to classify colorectal tumours into immune-hot, immune-cold, and intermediate phenotypes using transcriptomic profiles from TCGA. The final dataset includes 403 patients across two independent cohorts (TCGA-COAD n=294, TCGA-READ n=109). The pipeline integrates preprocessing, ssGSEA/ESTIMATE-based immune scoring, dimensionality reduction, unsupervised clustering, hypothesis testing (Welch t-test, ANOVA, Kruskal-Wallis, Mann-Whitney U, chi-square), machine learning classification, survival analysis, and external validation. Cross-validated model performance in the discovery cohort reached balanced accuracy of 0.785, while external validation achieved balanced accuracy of 0.744. Differential expression and survival analyses further support clinically meaningful immune stratification.

and often exhibit improved response to checkpoint blockade therapies. Immune-cold tumours, in contrast, show poor immune infiltration and weaker immunotherapy benefit. This project addresses a practical clinical informatics question: can we predict tumour immune phenotype from routine bulk RNA-seq data using a statistically rigorous and reproducible analysis stack?

2. Materials and Methods

2.1 Cohorts and Data Curation

Data were downloaded from GDC as STAR count RNA-seq

Table 1. Final cohort composition after preprocessing and immune-label assignment.

2.2 Preprocessing and Harmonization

Raw counts were converted with log2(CPM + 1), and genes were filtered at >=1 CPM in >=20% of samples. Top-variance genes (K=5000) were used for machine learning. Gene-symbol intersection harmonization between cohorts yielded 16,453 common genes, ensuring consistent feature spaces for downstream analyses.

2.3 Statistical and Machine Learning Pipeline

Immune labels were generated by ssGSEA with ESTIMATE/Bindea signatures and quantile-based stratification (top third: Hot, bottom third: Cold, middle: Intermediate). Dimensionality reduction included PCA, ICA,

stratified 5-fold CV.

3. Results

3.1 Representation Learning and Immune Separation

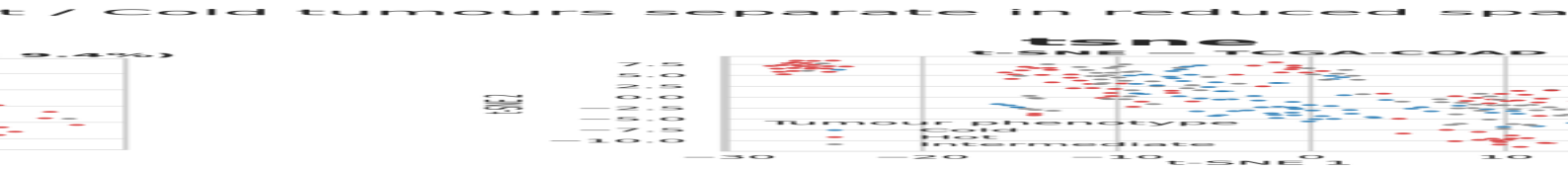

Figure 1. PCA/t-SNE/UMAP composite for TCGA-COAD showing partial immune-phenotype separation.

3.2 Differential Expression and Hypothesis Testing

In TCGA-COAD (16,723 genes tested), significant genes after BH correction were: t-test = 8198, ANOVA = 10369, Kruskal-Wallis = 10280, Mann-Whitney U = 8362. Applying combined BH + |log2FC| > 1 criteria yielded 2,269 high-confidence DE genes.

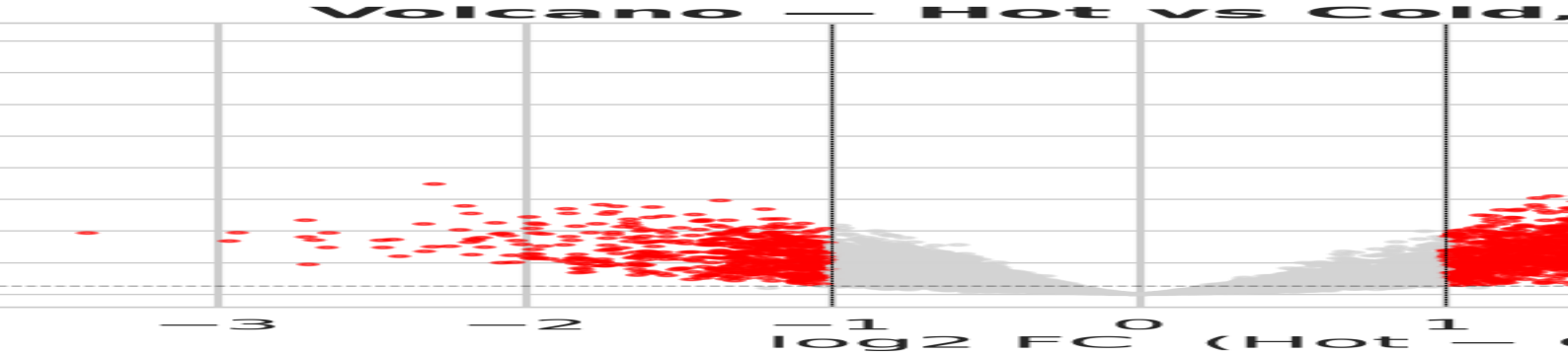

Figure 2. Volcano plot (Hot vs Cold, TCGA-COAD) with BH-adjusted significance thresholds.

3.3 Classification Performance

Model	Balanced Acc.	Macro F1
RandomForest	0.785	0.786
SVM-linear	0.785	0.784
Logistic	0.752	0.748
SVM-RBF	0.735	0.736
XGBoost	0.704	0.699
kNN-5	0.648	0.645

Table 2. Five-fold cross-validation performance in TCGA-COAD.

Figure 3. Model comparison in discovery cohort (balanced accuracy, 5-fold CV).

3.4 External Validation and Survival

Using the best discovery model (RandomForest), external testing on TCGA-READ reached accuracy = 0.743, balanced accuracy = 0.744, macro-F1 = 0.742 (n = 109). Kaplan-Meier

analysis in TCGA-COAD gave log-rank p = 0.081.

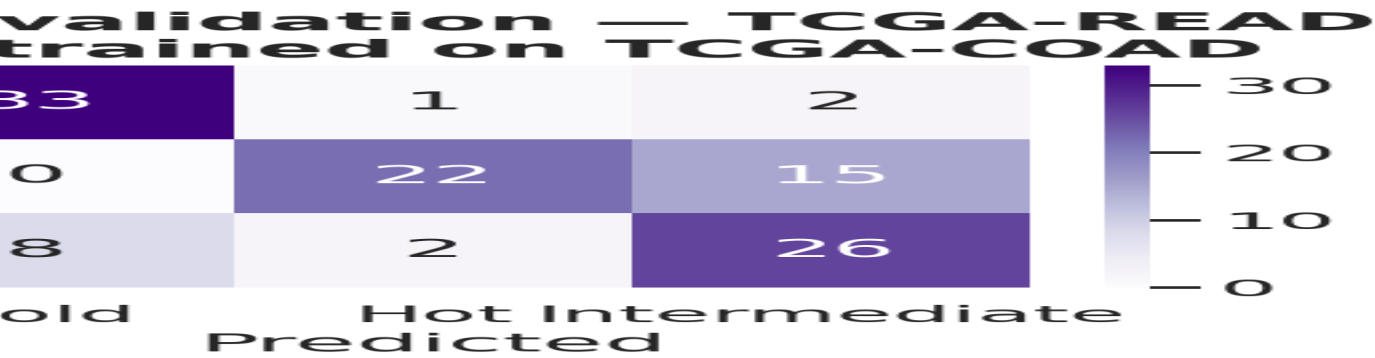

Figure 4. External validation confusion matrix (COAD-trained model evaluated on READ).

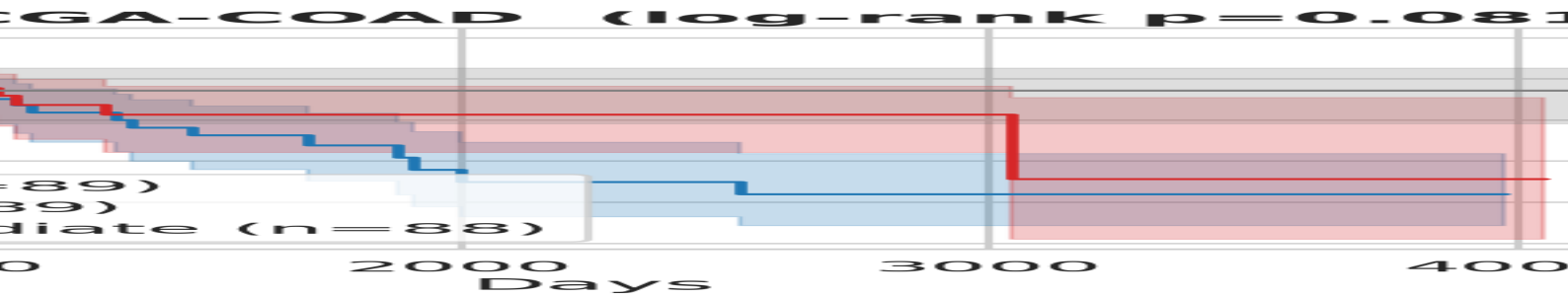

Figure 5. Kaplan-Meier survival curves for Hot/Cold/Intermediate in TCGA-COAD.

4. Discussion

The analysis confirms that a compact two-cohort design (403 total patients) is sufficient for a rigorous BT3041-grade translational bioinformatics study. The strongest performance was obtained by Random Forest and linear SVM, while XGBoost did not outperform simpler baselines in this feature regime. Statistical testing consistently identified robust immune-associated expression differences, supporting biological interpretability in addition to predictive performance.

5. Conclusion

This project delivers a full reproducible pipeline for immune-phenotype classification in colorectal tumours, covering all major statistical and machine-learning topics in the BT3041 syllabus. Future extension can include a cross-platform GEO cohort for stricter external generalization testing.

Code and Reproducibility

GitHub repository: https://github.com/Aryan7865/cold-hot-tumour-classification

All figures and tables in this report are auto-generated from scripts 01–10 and stored under outputs/.